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Drone/UAV Background

1.1 Introduction

Floods are among the most common natural disasters in Malaysia, especially in areas such as Kuala Lumpur, Kelantan, and Johor. Heavy rainfall, poor drainage systems, and climate change have contributed to the increasing number of flood incidents in recent years. Floods often cause major damage to property, transportation systems, and public safety.

During severe flood situations, rescue teams face many difficulties when trying to locate and assist stranded victims. Roads may become blocked, water levels may rise rapidly, and communication systems can be disrupted. These conditions make rescue operations slower and more dangerous for both victims and emergency responders.

Thus to address these issues, Unmanned Aerial Vehicles (UAV) or drones can be used as a solution as they possess unique capabilities which are able to enhance traditional efforts of rescue especially in accessing hazardous or unreachable areas.

1.2 Drone/UAV Overview

A UAV or drone is aircraft equipment which is able to be controlled remotely by either real-time human operators or through pre-programmed flight plans managed by computers or software for the UAV. The term 'drone' may be used to describe unmanned remotely piloted aircraft systems (RPAS).

Based on the aerodynamic design of UAVs they can be categorized into two main categories of UAV:

- Fixed-Wing UAVs - UAVs which resemble standard aeroplanes allowing a long flight range and endurance thus suitable for surveillances of large areas
- Rotary-Wing UAVs - UAVs which resemble quadcopters or hexacopters which allows vertical take off and landing (VTOL), good manoeuvrability and hover movements.

For disaster responses such as flood rescue, Rotary-Wing UAV would be preferred due to its good manoeuvrability in potentially tight environments, its ability to hover in place and VTOL ability.

1.3 Key Components of UAV

In order to ensure a UAV system is able to perform well functionally especially in rescue or surveillance operations, several core components are included to allow flight, navigation and data collection:

- Flight Controller - The computer/controller to process sensor data and allow control of the UAV
- GPS module - real-time positional data to allow autonomous navigation, following waypoints or returning to original site of takeoff
- Payload - mission-specific equipment such as cameras, thermal imaging sensors, speakers or other delivery functionality

1.4 Drone/UAV Technology in Disaster Response

In recent years, drones have become increasingly useful in disaster management and emergency response. Drones can access flooded or dangerous areas more quickly compared to traditional rescue vehicles. They are also capable of providing aerial monitoring and collecting real-time information during emergencies.

The use of drones can help:

- Improve search efficiency
- Reduce rescue response time
- Increase safety for rescue teams
- Monitor large disaster areas more effectively

Because of these advantages, drones are becoming an important technology in modern rescue operations. In several countries, many have integrated UAVs as part of their disaster response protocol.

1.5 UAV Limitations and Challenges

Despite many advantages in UAVs to enhance rescue operations, there are also operational and technical challenges which should be considered:

- Limited flight duration - Many commercially available drones are limited to 30 minutes - 1 hour of flight time which may be a restriction for long operations.
- Weather sensitive - many external factors such as strong winds, rain or low visibility may affect overall performance of UAV.
- Interference - Due to the nature of disasters, there may be damaged infrastructure or electromagnetic interference which can disrupt control link of UAV
- Payload limits - limitations in amount of supplies carried by UAV due to lifting capacity and reducing battery life.